

Python - Copy Sets

Python Copy Sets

Copying sets in Python refers to creating a new **set** that contains the same elements as an existing set. Unlike simple variable assignment, which creates a reference to the original set, copying ensures that changes made to the copied set do not affect the original set, and vice versa.

There are different methods for copying a set in Python, including using the `copy()` method, the `set()` function or set comprehension.

Copy Sets Using the `copy()` Method

The `copy()` method in set class is used to create a shallow copy of a set object.

A shallow copy means that the method creates a new collection object, but does not create copies of the objects contained within the original collection. Instead, it copies the references to these objects.

Therefore, if the original collection contains mutable objects (like lists, dictionaries, or other sets), modifications to these objects will be reflected in both the original and the copied collections.

Syntax

Following is the syntax of the `copy()` method –

```
set.copy()
```

Return Value

The `copy()` method returns a new set which is a shallow copy of existing set.

Example

In the following example, we are creating a copy of the set "lang1" and storing it in "lang2", then retrieving both sets and their memory addresses using id().

After adding an element to "lang1", we retrieve both sets and their memory addresses again to show that "lang1" and "lang2" are independent copies –

```
lang1 = {"C", "C++", "Java", "Python"}
print ("lang1: ", lang1, "id(lang1): ", id(lang1))
lang2 = lang1.copy()
print ("lang2: ", lang2, "id(lang2): ", id(lang2))
lang1.add("PHP")
print ("After updating lang1")
print ("lang1: ", lang1, "id(lang1): ", id(lang1))
print ("lang2: ", lang2, "id(lang2): ", id(lang2))
```

Output

This will produce the following output –

```
lang1: {'Python', 'Java', 'C', 'C++'} id(lang1): 2451578196864
lang2: {'Python', 'Java', 'C', 'C++'} id(lang2): 2451578197312
After updating lang1
lang1: {'Python', 'C', 'C++', 'PHP', 'Java'} id(lang1): 2451578196864
lang2: {'Python', 'Java', 'C', 'C++'} id(lang2): 2451578197312
```

Copy Sets Using the set() Function

The Python set() function is used to create a new set object. It takes an iterable as an argument and convert it into a set, removing any duplicate elements in the process. If no argument is provided, it creates an empty set.

We can copy set using the set() function by passing the original set as an argument to the set() constructor. This creates a new set that contains all the elements of the original set, ensuring that any modifications to the new set do not affect the original set.

Example

In this example, we are creating a copy of "original_set" using the set() function and storing it in "copied_set" –

```
# Original set
original_set = {1, 2, 3, 4}
# Copying the set using the set() function
copied_set = set(original_set)
print("copied set:", copied_set)

# Demonstrating that the sets are independent
copied_set.add(5)
print("copied set:", copied_set)
print("original set:", original_set)
```

Output

Following is the output of the above code –

```
copied set: {1, 2, 3, 4}
copied set: {1, 2, 3, 4, 5}
original set: {1, 2, 3, 4}
```

Copy Sets Using Set Comprehension

Set comprehension is a concise way to create sets in Python. It is used to generate a new set by iterating over an iterable and optionally applying conditions to filter elements. The syntax is similar to list comprehension but with curly braces {} instead of square brackets [] –

```
{expression for item in iterable if condition}
```

We can copy sets using set comprehension by iterating over the elements of the original set and directly creating a new set with those elements.

Example

In the example below, we create an original set named "original_set", then copy it using set comprehension into "copied_set" –

```
# Original set
original_set = {1, 2, 3, 4, 5}

# Copying the set using set comprehension
copied_set = {x for x in original_set}
print("Copied set:", copied_set)
```

Output

Output of the above code is as shown below –

```
Copied set: {1, 2, 3, 4, 5}
```

TOP TUTORIALS

[Python Tutorial](#)

[Java Tutorial](#)

[C++ Tutorial](#)

[C Programming Tutorial](#)

[C# Tutorial](#)

[PHP Tutorial](#)

[R Tutorial](#)

[HTML Tutorial](#)

[CSS Tutorial](#)

[JavaScript Tutorial](#)

[SQL Tutorial](#)

TRENDING TECHNOLOGIES

[Cloud Computing Tutorial](#)

[Amazon Web Services Tutorial](#)

[Microsoft Azure Tutorial](#)

[Git Tutorial](#)

[Ethical Hacking Tutorial](#)

[Docker Tutorial](#)

[Kubernetes Tutorial](#)

[DSA Tutorial](#)

[Spring Boot Tutorial](#)

[SDLC Tutorial](#)

[Unix Tutorial](#)

CERTIFICATIONS

[Business Analytics Certification](#)

[Java & Spring Boot Advanced Certification](#)

[Data Science Advanced Certification](#)



[Chapters](#) ▾

[Categories](#)

[Artificial Intelligence And Machine Learning](#)

[DevOps Certification](#)

[Game Development Certification](#)

[Front-End Developer Certification](#)

[AWS Certification Training](#)

[Python Programming Certification](#)

COMPILERS & EDITORS

[Online Java Compiler](#)

[Online Python Compiler](#)

[Online Go Compiler](#)

[Online C Compiler](#)

[Online C++ Compiler](#)

[Online C# Compiler](#)

[Online PHP Compiler](#)

[Online MATLAB Compiler](#)

[Online Bash Terminal](#)

[Online SQL Compiler](#)

[Online Html Editor](#)

[ABOUT US](#) | [OUR TEAM](#) | [CAREERS](#) | [JOBS](#) | [CONTACT US](#) | [TERMS OF USE](#) |

[PRIVACY POLICY](#) | [REFUND POLICY](#) | [COOKIES POLICY](#) | [FAQ'S](#)



Tutorials Point is a leading Ed Tech company striving to provide the best learning material on technical and non-technical subjects.

© Copyright 2026. All Rights Reserved.